

LDV_E10

Gasoline-Ethanol Blend (E10) Light Duty Vehicle

Inputs / Outputs

dir	comm	unit	group
Input	GSL	PJ	i
Input	BIO	PJ	i
Output	PLDVHWY	MPK _m	o
Output	PLDVCTY	MPK _m	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
2	BIO	NA	NA	0.1
3	PLDVHWY	679.995	2.03	0.4
4	PLDVCTY	502.401	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
cap unit	1000 Vehicles
act unit	million km, city
cost unit	MUSD

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	GSL + BIO	4.30	2.45	1.47	54.70
City	GSL + BIO	5.82	1.81	1.99	40.41

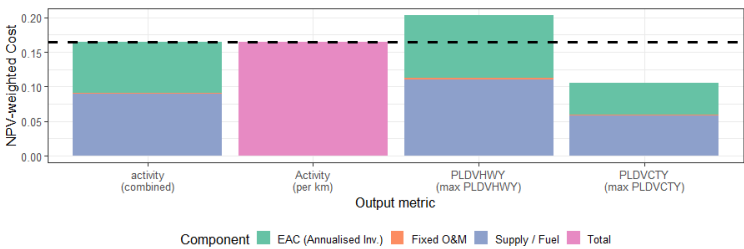
Vehicle Parameters

Fuels	GSL, BIO	Primary fuel	GSL
Efficiency (highway)	4.3 L/100km	Efficiency (city)	5.82 L/100km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Blend fuel	BIO (max 10%)	Vehicle cost	8,414 USD
Annual fixed cost	25 USD/yr	Lifetime	16 yr

Costs [MUSD]

Fixed O&M	0.02
Investment cost	8.41

Levelized Cost NPV = 0.17 [MUSD]



Schematic

